

MINOR PROJECT REPORT

TITLE PAGE

MyCircle – A Local Student Opportunity & Networking Platform

A Minor Project Report submitted in partial fulfillment of the requirements for the award of the degree of MASTER OF COMPUTER APPLICATIONS

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MCA 2nd Semester

Department of Computer Science and Information Technology

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CERTIFICATE OF APPROVAL

This is to certify that the Minor Project Report entitled "MyCircle – A Local Student Opportunity & Networking Platform" submitted by [STUDENT_NAME], MCA 2nd Semester student at Guru Ghasidas Vishwavidyalaya, Bilaspur (C.G.), is a bonafide record of the project work carried out by him/her under my supervision and guidance.

The project has been completed in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications and is fit for submission to the University.

Date: _____

Place: Bilaspur (C.G.)

(Signature of External Examiner)

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ABSTRACT

MyCircle – A Local Student Opportunity & Networking Platform

The landscape of higher education in India has undergone significant transformation in recent years, with students increasingly seeking opportunities that align with their local college ecosystems. However, there exists a substantial gap in the current digital ecosystem - while professional networking platforms like LinkedIn serve the corporate world, no dedicated platform exists specifically for college students to discover and share local opportunities, connect with peers, and build meaningful professional networks.

MyCircle is a comprehensive local student opportunity and networking platform developed using the MERN stack (MongoDB, Express.js, React.js, Node.js) that bridges this critical

gap. The platform enables students to post and discover local opportunities including internships, part-time jobs, freelance gigs, campus events, hackathons, and study groups. Students can build verified profiles linked to their institution, follow their peers, message each other in real-time, and collaborate on projects.

The MyCircle application features a modern, responsive user interface built with React and Tailwind CSS, providing seamless experience across desktop and mobile devices. Security is ensured through JWT-based authentication, bcrypt password hashing, and role-based access control.

Keywords: MERN Stack, Student Networking, Local Opportunities, Web Platform, MongoDB, React.js, Node.js

CHAPTER 1: INTRODUCTION

1.1 Introduction to the Project

The digital revolution has transformed how we connect, collaborate, and seek opportunities in unprecedented ways. While global platforms like LinkedIn, Facebook, and Twitter have revolutionized professional and social networking at a macro level, a significant gap remains unaddressed at the local level - specifically within college and university ecosystems where students spend their most formative years.

MyCircle emerges as a purpose-built solution designed specifically for local college and university student communities. It represents a comprehensive web platform that enables students to post and discover local opportunities including internships, part-time jobs, freelance gigs, campus events, hackathons, workshops, and study groups within their immediate college circle.

1.2 Problem Statement

The problem that MyCircle addresses is multifaceted and deeply rooted in the current landscape of how college students access opportunities and connect with each other.

Despite the proliferation of digital platforms in our daily lives, students in local colleges and universities face significant challenges in accessing relevant opportunities within their immediate geographic and institutional circles.

1.3 Motivation

The motivation behind building MyCircle stems from both personal experiences as a student and a genuine recognition of the gap in the current digital ecosystem.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Platforms Analysis

LinkedIn - Professional networking but not student-focused

Internshala - Focuses only on internships, missing broader student opportunities

WhatsApp - Communication but lacks structured opportunity management

College Portals - Often outdated user experience

2.2 Comparative Analysis

Platform	Target Audience	Local Focus	Opportunity Posting	Cost
LinkedIn	Professionals	National	Yes	Free + Premium
Internshala	Job Seekers	National	Yes	Free + Premium
WhatsApp	General	Local	No	Free
MyCircle	Students	Local	Yes	Free

CHAPTER 3: SYSTEM REQUIREMENTS

3.1 Functional Requirements

- User Registration and Login (JWT Auth)
- Student Profile Creation and Management
- Post Opportunities (internships, gigs, jobs)
- Browse and Search Opportunities
- Create and Join Study Groups
- Post and Discover Campus Events
- Follow/Unfollow System
- Real-time Notifications
- Messaging System
- Admin Panel

3.2 Non-Functional Requirements

- Performance: Page load under 3 seconds
- Security: JWT, bcrypt, HTTPS
- Scalability: Horizontal scaling support
- Availability: 99% uptime target

3.3 Technology Stack

Software	Version	Purpose
Node.js	18.x	JavaScript runtime
Express.js	4.x	Web framework
React.js	18.x	Frontend UI

Software	Version	Purpose
MongoDB	6.x	Database
Vite	5.x	Build tool
Tailwind CSS	3.x	Styling
JWT	Latest	Authentication
bcrypt	Latest	Password hashing

CHAPTER 4: SYSTEM DESIGN

4.1 System Architecture

MyCircle employs a three-tier architecture:

- Client Tier: React.js SPA built with Vite
- API Tier: Node.js Express.js server
- Data Tier: MongoDB database

4.2 Database Design (Collections)

Users Collection:

- `_id`, email, password, name, college, course, year, bio, profileImage, role, followers, following

Opportunities Collection:

- `_id`, title, type, description, postedBy, location, requirements, applicationLink, deadline

Events Collection:

- `_id`, title, description, date, venue, organizer, registrationLink

StudyGroups Collection:

- `_id`, `name`, `subject`, `description`, `admin`, `members`

Messages Collection:

- `_id`, `sender`, `receiver`, `content`, `timestamp`, `read`

Notifications Collection:

- `_id`, `recipient`, `type`, `message`, `isRead`, `createdAt`

CHAPTER 5: TECHNOLOGY STACK

5.1 MongoDB

MongoDB is a leading NoSQL database that stores data in flexible, JSON-like documents. The document model aligns naturally with JavaScript object structures.

5.2 Express.js

Express.js is a minimalist web framework for Node.js that provides robust features for building web applications and APIs.

5.3 React.js

React.js is a JavaScript library for building user interfaces through its component-based architecture.

5.4 Node.js

Node.js provides the JavaScript runtime that executes MyCircle's server-side code.

5.5 Vite

Vite serves as MyCircle's frontend build tool and development server.

5.6 Tailwind CSS

Tailwind CSS provides MyCircle's styling through its utility-first approach.

5.7 JWT

JWT provides MyCircle's authentication through stateless token-based security.

CHAPTER 6: IMPLEMENTATION

6.1 User Registration and Authentication

The authentication system represents the foundation of MyCircle's security model. Users register with full name, email, password, college, year, and course.

6.2 Student Profile Page

Profile page serves as students' personal showcase with edit functionality and profile picture upload via Cloudinary.

6.3 Opportunities Feed

Main discovery interface showing all opportunities in a scrollable card layout with filtering options.

6.4 Post an Opportunity

Create forms with title, type, description, location, requirements, application link, and deadline.

6.5 Campus Events

Event listings with registration features.

6.6 Study Groups

Create, browse, join, and leave study groups.

6.7 Follow/Unfollow System

Enables personalized feeds and connection building.

6.8 Messaging

Direct communication between users.

6.9 Notifications

Alerts for new followers, opportunity posts, and event reminders.

6.10 Admin Panel

Moderation and statistical features.

CHAPTER 7: TESTING

7.1 Testing Strategy

MyCircle employs multi-level testing: Unit, Integration, System, and User Acceptance Testing.

7.2 API Test Results

Test Case	Endpoint	Method	Status
Register	/auth/register	POST	Pass
Login	/auth/login	POST	Pass
Create Opportunity	/opportunities	POST	Pass
Follow User	/user/follow/:id	POST	Pass
Create Event	/events	POST	Pass

CHAPTER 8: UI SCREENSHOTS

8.1 Login / Register Page

Centered card on dark gradient background.

8.2 Home / Opportunity Feed

Card layout with color-coded type badges.

8.3 Profile Page

Profile picture, name, college, and bio.

8.4 Post Opportunity Form

Modal form with validation.

CHAPTER 9: LIMITATIONS

9.1 Technical Limitations

- No mobile application (web only)
- Basic recommendation algorithm
- Polling for messaging (not WebSocket)
- Basic search with regex

9.2 Platform Limitations

- Free deployment tier limits
- Storage limits on free tiers

9.3 Security Limitations

- No 2FA
 - JWT in localStorage
 - Rate limiting not fully implemented
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CHAPTER 10: FUTURE SCOPE

10.1 AI-Powered Recommendations

Machine learning for personalized feeds.

10.2 Mobile Application

React Native for native mobile app.

10.3 Multi-College Expansion

Additional verification infrastructure.

10.4 Advanced Search

Elasticsearch integration.

10.5 Gamification

Points, badges, and leaderboards.

10.6 Mentorship Module

Connecting students with mentors.

10.7 Premium Features

Monetization options.

CHAPTER 11: CONCLUSION

11.1 Summary

MyCircle stands as a comprehensive local student opportunity and networking platform developed using the MERN stack.

11.2 Learning Outcomes

Full-stack development, REST API design, JWT authentication, database modeling, React state management.

11.3 Impact

Centralized opportunity discovery, verified student profiles, community building, event discovery.

11.4 Closing Statement

MyCircle represents how digital platforms can support local educational communities.

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